Classical adversary bounds lift to randomized communication complexity with a constant-size versatile gadget

Anurag Anshu, Shalev Ben-David, Srijita Kundu

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Editorial note. Editable transcription of corresponding formally published CC BY classical lifting statement/proof, using nonexclusive native only as private aid. Public body blanks all excluded native ranges and comments while preserving locator lines; no whole unselected native file is publicly referenced. Author math macros flattened without input/include; unused custom bbold alphabet omitted. Native global theorem offset restored for selected background, and comparison mentions of omitted query-relations chapter rendered as accurate originalSection6. Quantum conditional lifting and separate query-relation proofs not imported into this classical record. All production/extraction/validation solely on Yupeng; no local compilation.

Allowed standard prerequisites (editorial).

- Randomized communication with public randomness, classical adversary weight schemes, Hellinger distance and conditional information identities. Complete originalSection2 models/measure definitions and standard cited information facts retained; source-local versatile gadget/sampling and complete lifting proof included

Theorem 1 (Original Theorem 1)**.** *There is an explicitly given function* $G : \mathcal{X} \times \mathcal{Y} \to \{0,1\}$ *such that for any (possibly partial) Boolean function or relation* f*,*

$$\mathrm{RCC}(f \circ G) = \Omega(\mathrm{CAdv}(f)).$$

Here $\mathrm{RCC}(f \circ G)$ denotes the randomized communication complexity of $f \circ G$ with shared randomness.

2 Preliminaries

2.1 Distance & information measures

We define all the distance and information measures for quantum states. The classical versions can be obtained by making the corresponding registers classical.

The ℓ_1 distance between two quantum states ρ and σ is defined as

$$\|\rho - \sigma\|_1 = \operatorname{Tr} \sqrt{(\rho - \sigma)^\dagger (\rho - \sigma)}.$$

The entropy of a quantum state ρ_A on register A is defined as

$$H(A)_\rho = -\operatorname{Tr}(\rho \log \rho).$$

For a state ρ_{AB} on registers AB, the conditional entropy of A given B is

$$H(A|B)_\rho = H(AB)_\rho - H(B)_\rho.$$

Conditional entropy satisfies the following continuity bound [4]: if ρ and σ on registers AB satisfy $\|\rho - \sigma\|_1 \leq \epsilon$, then

$$|H(A|B)_\rho - H(A|B)_\sigma| \leq 4\epsilon \log |A| + 2h(\epsilon)$$

where $h(.)$ is the binary entropy function. For ρ_{ABC}, we define the mutual information and conditional mutual information as

$$I(A:B)_\rho = H(A)_\rho - H(A|B)_\rho \qquad I(A:B|C) = H(A|C)_\rho - H(A|BC)_\rho.$$

Mutual information satisfies

$$0 \leq I(A:B|C)_\rho \leq \min\{\log |A|, \log |B|\}$$

and the chain rule

$$I(A:BC)_\rho = I(A:B)_\rho + I(A:C|B)_\rho.$$

2.2 Query complexity

In query complexity, the primary object of study are Boolean functions, which are functions $f\colon \{0,1\}^n \to \{0,1\}$ where n is a positive integer. Often, we will actually study partial Boolean functions, which are defined on only a subset of $\{0,1\}^n$. We will use $\mathrm{Dom}(f)$ to denote the domain of f; this is a subset of $\{0,1\}^n$.

For a (possibly partial) Boolean function f, we use $\mathrm{D}(f)$, $\mathrm{R}(f)$, and $\mathrm{Q}(f)$ to denote its deterministic query complexity, randomized query complexity (to bounded error), and quantum query complexity (to bounded error), respectively. For the definition of these measures, see [9], though we won't use these definitions in this work.

2.2.1 Block sensitivity and its variants

We will use the following definitions.

Block notation. For a Boolean string $x \in \{0,1\}^n$ and a set $B \subseteq [n]$, we let x^B denote the string with the bits in B flipped; that is, $x^B_i = x_i$ for all $i \notin B$ and $x^B_i = 1 - x_i$ for all $i \in B$. The set B is called a *block*.

Sensitive block. For a (possibly partial) Boolean function f on n bits and an input $x \in \mathrm{Dom}(f)$, we say that a set $B \subseteq [n]$ is a *sensitive block* for x (with respect to f) if $x^B \in \mathrm{Dom}(f)$ and $f(x^B) \neq f(x)$.

Block sensitivity. The *block sensitivity* of a string $x \in \{0,1\}^n$ with respect to a (possibly partial) Boolean function f satisfying $x \in \mathrm{Dom}(f)$ is the maximum integer k such that there are k blocks $B_1, B_2, \ldots, B_k \subseteq [n]$ which are all sensitive for x and which are all disjoint. This is denoted $\mathrm{bs}(x, f)$.

Block sensitivity of a function. The *block sensitivity* of a (possibly partial) Boolean function f is the maximum value of $\mathrm{bs}(x, f)$ over $x \in \mathrm{Dom}(f)$. This is denoted $\mathrm{bs}(f)$. Block sensitivity was originally introduced by Nisan [15], and is discussed in the survey by Buhrman and de Wolf [9].

Fractional block sensitivity. The *fractional block sensitivity* of a string $x \in \{0,1\}^n$ with respect to a (possibly partial) Boolean function f satisfying $x \in \mathrm{Dom}(f)$ is the maximum possible sum of weights $\sum_B w_B$, where the weights $w_B \geq 0$ are assigned to each sensitive block of x and must satisfy $\sum_{B:i \in B} w_B \leq 1$ for all $i \in [n]$. This is denoted by $\mathrm{fbs}(x, f)$. The fractional block sensitivity of a function f, denoted $\mathrm{fbs}(f)$, is the maximum value of $\mathrm{fbs}(x, f)$ over $x \in \mathrm{Dom}(f)$. Fractional block sensitivity was defined by [2], but see also [12].

Critical block sensitivity. For a (possibly partial) Boolean function f, we say that a total Boolean function f' is a *completion* of f if $f'(x) = f(x)$ for all $x \in \mathrm{Dom}(f)$. The *critical block sensitivity* of f, denoted $\mathrm{cbs}(f)$, is defined as

$$\min_{f'} \max_{x \in \mathrm{Dom}(f)} \mathrm{bs}(x, f'),$$

where the minimum is taken over completions f' of f. This measure was defined by [11]. It equals $\mathrm{bs}(f)$ for total functions, but may be larger for partial functions.

Critical fractional block sensitivity. For a (possibly partial) Boolean function f, we define its *critical fractional block sensitivity*, denoted $\mathrm{cfbs}(f)$, as

$$\min_{f'} \max_{x \in \mathrm{Dom}(f)} \mathrm{cfbs}(x, f'),$$

where the minimum is taken over completions f' of f. This measure has not previously appeared in the literature.

2.2.2 Adversary bounds

Positive adversary bound. For a (possibly partial) Boolean function f, we define the positive-weight adversary bound, denoted $\mathrm{Adv}(f)$, as the minimum of the following program. We will have one non-negative weight $q(x, i)$ for each $x \in \mathrm{Dom}(f)$ and each $i \in [n]$. We call such a weight scheme feasible if, for all $x, y \in \mathrm{Dom}(f)$ with $f(x) \neq f(y)$, we have

$$\sum_{i : x_i \neq y_i} \sqrt{q(x, i) q(y, i)} \geq 1.$$

Then $\mathrm{Adv}(f)$ is defined as the minimum of $\max_{x \in \mathrm{Dom}(f)} \sum_{i \in [n]} q(x, i)$ over feasible weight schemes $q(\cdot, \cdot)$. A different version of the positive-weight adversary bound was defined in [6], though the version we've currently defined appears in [13] and [16] (in the latter, our definition is equivalent to $\mathrm{MM}(f)$).

Classical adversary bound. For a (possibly partial) Boolean function f, we define the classical adversary bound, denoted $\mathrm{CAdv}(f)$, as the minimum of the following program. We will have one non-negative weight $q(x,i)$ for each $x \in \mathrm{Dom}(f)$ and each $i \in [n]$, as before. We call such a weight scheme feasible if, for all $x, y \in \mathrm{Dom}(f)$ with $f(x) \neq f(y)$, we have

$$\sum_{i:x_i \neq y_i} \min\{q(x,i), q(y,i)\} \geq 1.$$

Then $\mathrm{CAdv}(f)$ is defined as the minimum of $\max_{x \in \mathrm{Dom}(f)} \sum_{i \in [n]} q(x,i)$ over feasible weight schemes $q(\cdot,\cdot)$. Observe that this definition is the same as that of $\mathrm{Adv}(f)$, except that the feasibility constraint sums up the minimum of $q(x,i)$ and $q(y,i)$ instead of the geometric mean. This feasibility constraint is harder to satisfy, and hence we have $\mathrm{CAdv}(f) \geq \mathrm{Adv}(f)$. A different version of the classical adversary was defined in [1], though the version we've currently defined appears in [13] and [5] (in the latter, our definition is equivalent to $\mathrm{CMM}(f)$).

Singleton adversary bound. [3] introduced a simplified version of the quantum adversary bound, which we denote $\mathrm{Adv}_1(f)$. As in the other adversaries, this will be the minimum over a program that has one non-negative weight $q(x,i)$ for each pair of input $x \in \mathrm{Dom}(f)$ and index $i \in [n]$. The objective value will once again be $\max_{x \in \mathrm{Dom}(f)} \sum_{i \in [n]} q(x,i)$. The only difference is the constraints: instead of placing a constraint for each $x, y \in \mathrm{Dom}(f)$ with $f(x) \neq f(y)$, we only place this constraint for such x, y that have Hamming distance exactly 1. Observe that this is a relaxation of the constraint in the definition of $\mathrm{Adv}(f)$, and hence $\mathrm{Adv}_1(f) \leq \mathrm{Adv}(f)$ for all (possibly partial) Boolean functions f.

2.3 A generalization to relations

So far, we've defined our query measures for partial Boolean functions. However, in many cases we will be interested in studying *relations*, which are a generalization of partial Boolean functions.

In query complexity, a *relation* is a subset of $\{0,1\}^n \times \Sigma$, where Σ is some finite output alphabet. We will equate a relation $f \subseteq \{0,1\}^n \times \Sigma$ with a function that maps $\{0,1\}^n$ to subsets of Σ, so that for a string $x \in \{0,1\}^n$, the notation $f(x)$ denotes $\{\sigma \in \Sigma : (x,\sigma) \in f\}$. An algorithm which computes a relation f to error ϵ must have the guarantee that for inputs $x \in \{0,1\}^n$, the algorithm outputs a symbol in $f(x)$ with probability at least $1 - \epsilon$.

Relations are generalizations of partial functions. This is because we can represent a partial function f with domain $\mathrm{Dom}(f) \subseteq \{0,1\}^n$ by a relation f' such that $f'(x) = \{f(x)\}$ for $x \in \mathrm{Dom}(f)$ and $f'(x) = \{0,1\}$ for $x \notin \mathrm{Dom}(f)$. In other words, the relational version f' of the partial function f will accept all input strings (it will be a total function), but it will consider every output symbol to be valid when given an input not in $\mathrm{Dom}(f)$. This essentially makes the inputs not in $\mathrm{Dom}(f)$ become trivial, and hence makes the relation f' intuitively equivalent to the partial function f.

We will generalize several of our query measures to relations.

Critical (fractional) block sensitivity. The original definition of $\mathrm{cbs}(f)$ from [11] actually defined it for relations. We say that a total function $f' \colon \{0,1\}^n \to$

Σ is a *completion* of a relation $f \subseteq \{0,1\}^n \times \Sigma$ if $(x, f'(x)) \in f$ for all $x \in \{0,1\}^n$. In other words, f' is a completion if it gives a fixed, valid output choice for each input to f. Next, we say an input $x \in \{0,1\}^n$ is *critical* if it has a unique valid output symbol in f; that is, if $|f(x)| = 1$. We let $\mathrm{crit}(f)$ denote the set of all critical inputs to f. (Note that if f is the relational version of a partial function, then $\mathrm{crit}(f)$ is equal to the domain of the partial function.) We then define

$$\mathrm{cbs}(f) \coloneqq \min_{f'} \max_{x \in \mathrm{crit}(f)} \mathrm{bs}(x, f')$$

$$\mathrm{cfbs}(f) \coloneqq \min_{f'} \max_{x \in \mathrm{crit}(f)} \mathrm{fbs}(x, f'),$$

where the minimizations are over completions f' of f. Observe that if f is the relational version of a partial function, these definitions match the previous ones.

Adversary bounds. The adversary bounds easily generalize to relations: both the positive adversary bound and the classical adversary bound will still be minimizations over weight schemes $q(x, i)$, with a non-negative weight assigned to each pair of input in $\{0,1\}^n$ and $i \in [n]$. The objective value to be minimized is the same as before: $\max_{x \in \{0,1\}^n} \sum_{i \in [n]} q(x, i)$. As for the constraints, we previously had one constraint for each pair of inputs x, y with $f(x) \neq f(y)$. For relations, we will replace this condition with the condition $f(x) \cap f(y) = \varnothing$ (that is, x and y have disjoint allowed-output-symbol sets). Hence the new constraint for $\mathrm{Adv}(f)$ becomes that for all pairs $x, y \in \{0,1\}^n$ with $f(x) \cap f(y) = \varnothing$, we have

$$\sum_{i : x_i \neq y_i} \sqrt{q(x,i) q(y,i)} \geq 1.$$

Similarly, the constraint for $\mathrm{CAdv}(f)$ is that for all pairs $x, y \in \{0,1\}^n$ with $f(x) \cap f(y) = \varnothing$, we have

$$\sum_{i : x_i \neq y_i} \min\{q(x,i), q(y,i)\} \geq 1,$$

and the constraint for $\mathrm{Adv}_1(f)$ is similar.

2.3.1 Degree measures

Degree of a function. For a (possibly partial) Boolean function f, we define its *degree* to be the minimum degree of a real polynomial p which satisfies $p(x) = f(x)$ for all $x \in \mathrm{Dom}(f)$ as well as $p(x) \in [0,1]$ for all $x \in \{0,1\}^n$. We denote this by $\deg(f)$.

Approximate degree. For a (possibly partial) Boolean function f, we define its *approximate degree* to error ϵ to be the minimum degree of a real polynomial p which satisfies $|p(x) - f(x)| \leq \epsilon$ for all $x \in \mathrm{Dom}(f)$ as well as $p(x) \in [0,1]$ for all $x \in \{0,1\}^n$. We denote this by $\widetilde{\deg}_\epsilon(f)$. When $\epsilon = 1/3$, we omit it and write $\widetilde{\deg}(f)$.

These measures are both defined and discussed in the survey by Buhrman and de Wolf [9]. We note that for partial functions, some authors do not include

the requirement that the polynomial approximating the function is bounded outside of the promise set. Without this requirement, one gets a smaller measure. In this work we will only use degree and approximate degree to refer to the bounded versions of these measures.

We also note that approximate degree can be *amplified*: if a polynomial p approximates a function f to error ϵ, then we can modify p to get a polynomial q which approximates f to error $\epsilon' < \epsilon$ and which has degree that is at most a constant factor larger than the degree of p (this constant factor will depend on ϵ and ϵ').

2.3.2 Known relationships between measures

See Figure 1 for a summary of relationships between these measures (for partial functions).

It is not hard to see that $\mathrm{bs}(f)$ is the smallest of the block sensitivity measures, and $\mathrm{cfbs}(f)$ is the largest. We know [2, 11] that $\mathrm{fbs}(f)$ and $\mathrm{cbs}(f)$ both lower bound $\mathrm{R}(f)$ for all (possibly partial) Boolean functions f; in Section 6 of the original publication, we show that $\mathrm{cfbs}(f)$ is also a lower bound.

We know [13, 16, 5] that $\mathrm{Q}(f) = \Omega(\mathrm{Adv}(f))$ and $\mathrm{R}(f) = \Omega(\mathrm{CAdv}(f))$. Although this it not ordinarily stated for relations, both lower bounds hold when f is a relation as well. In Section 6 of the original publication, we show that $\mathrm{CAdv}(f) = \Theta(\mathrm{cfbs}(f))$ for all partial functions f, and we also show that $\mathrm{CAdv}(f) = O(\mathrm{Adv}(f)^2)$ which holds for both partial functions and relations.

Approximate degree lower bounds quantum query complexity: $\mathrm{Q}(f) = \Omega(\widetilde{\deg}(f))$. It is known [7] that approximate degree is lower-bounded by $\sqrt{\mathrm{bs}(f)}$. Tal [17] showed that for total functions, $\widetilde{\deg}(f) = \Omega(\sqrt{\mathrm{fbs}(f)})$. In Section 6 of the original publication, we extend this result to partial functions, and also prove that $\widetilde{\deg}(f) = \Omega(\sqrt{\mathrm{cfbs}(f)})$.

In conclusion, $\mathrm{CAdv}(f)$ turns out to be the same as $\mathrm{cfbs}(f)$ for partial functions, and its square root lower bounds both $\mathrm{Adv}(f)$ and $\widetilde{\deg}(f)$, both of which are lower bounds on $\mathrm{Q}(f)$. Without taking square roots, $\mathrm{CAdv}(f)$ is a lower bound on $\mathrm{R}(f)$ but not on $\mathrm{Q}(f)$.

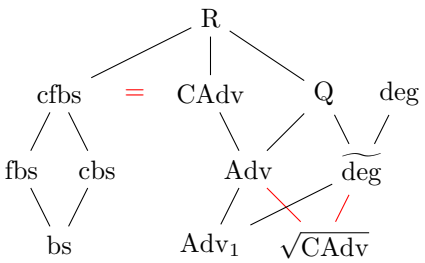


Figure 1: Relations between query complexity measures used in this work, applicable to partial functions. An upwards line from $\mathrm{M}_1(f)$ to $\mathrm{M}_2(f)$ means that $\mathrm{M}_1(f) = O(\mathrm{M}_2(f))$ for all (possibly partial) Boolean functions f. Red indicates new relationships proved in this work. We warn that some of these relationships are false for relations; in particular, CAdv may be strictly larger than cfbs and its square root may be incomparable to $\widetilde{\deg}$ for relations.

When we move from partial functions to relations, the measure $\mathrm{CAdv}(f)$ appears to get stronger in comparison to the other measures: it is strictly larger than $\mathrm{cfbs}(f)$, and appears to be incomparable to $\widetilde{\deg}(f)$ (though defining the latter for relations is a bit tricky, and we don't do so in this work).

2.4 Communication complexity

In the communication model, two parties, Alice and Bob, are given inputs $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ respectively, and in the most general case the task is to jointly compute a relation $f \subseteq \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$ by communicating with each other. In other words, on input (x, y), Alice and Bob must output a symbol $z \in \mathcal{Z}$ such that $(x, y, z) \in f$. Without loss of generality, we can assume Alice sends the first message, and Bob produces the output of the protocol.

In the classical randomized model, Alice and Bob are allowed to use shared randomness R (and also possibly private randomness R_A and R_B) in order to achieve this. The cost of a communication protocol Π, denoted by $\mathrm{CC}(\Pi)$ is the number of bits exchanged between Alice and Bob. The randomized communication complexity of a relation f with error ϵ, denoted by $\mathrm{R}_\epsilon^{\mathrm{CC}}(f)$, is defined as the minimum $\mathrm{CC}(\Pi)$ of a randomized protocol Π that computes f with error at most ϵ on every input.

Classical information complexity. The information complexity of a protocol with inputs X, Y according to μ, shared randomness R and transcript Π is given by

$$\mathrm{IC}(\Pi, \mu) = I(X : \Pi | YR)_\mu + I(Y : \Pi | XR)_\mu.$$

For any μ we have, $\mathrm{IC}(\Pi, \mu) \leq \mathrm{CC}(\Pi)$.

In a quantum protocol Π, Alice and Bob initially share an entangled state on registers $A_0 B_0$, and they get inputs x and y from a distribution μ. The global state at the beginning of the protocol is

$$|\Psi^0\rangle = \sum_{x,y} \sqrt{\mu(x,y)}\, |xxyy\rangle_{X\widetilde{X}Y\widetilde{Y}} \otimes |\Theta^0\rangle_{A_0 B_0}$$

where the registers $\widetilde{X}$ and $\widetilde{Y}$ purify X and Y and are inaccessible to either party. In the t-th round of the protocol, if t is odd, Alice applies a unitary $U_t : A'_{t-1} C_{t-1} \to A'_t C_t$, on her input, her memory register A'_{t-1} and the message C_{t-1} from Bob in the previous round (where $A'_0 = X A_0$ and C_0 is empty), to generate the new message C_t, which she sends to Bob, and new memory register A'_t. Similarly, if t is even, then Bob applies the unitary $U_t : B'_{t-1} C_{t-1} \to B'_t C_t$ and sends C_t to Alice. It is easy to see that $B'_t = B'_{t-1}$ for odd t, and $A'_t = A'_{t-1}$ for even t. We can assume that the protocol is safe, i.e., for all t, $A'_t = X A_t$ and $B'_t = Y B_t$, and U_t uses X or Y only as control registers. The global state at the t-th round is then

$$|\Psi^t\rangle = \sum_{x,y} \sqrt{\mu(x,y)}\, |xxyy\rangle_{X\widetilde{X}Y\widetilde{Y}} \otimes |\Theta^t\rangle_{A_t B_t C_t | xy}\,.$$

[14] (Proposition 9) showed that making a protocol safe does not decrease its QIC (defined below), so we shall often work with protocols of this form.

The quantum communication cost of a protocol Π, denoted by $\mathrm{QCC}(\Pi)$, is the total number of qubits exchanged between Alice and Bob in the protocol, i.e., $\sum_t \log |C_t|$. The quantum communication complexity of f with error ϵ, denoted by $\mathrm{Q}^{\mathrm{CC}}(f)$, is defined as the minimum $\mathrm{QCC}(\Pi)$ of a quantum protocol Π that computes f with error at most ϵ on every input.

Quantum information complexity. Given a quantum protocol Π as described above with classical inputs distributed as μ, its quantum information complexity is defined as

$$\mathrm{QIC}(\Pi,\mu) = \sum_{t \text{ odd}} I(\widetilde{X}\widetilde{Y} : C_t|YB'_t)_{\Psi^t} + \sum_{t \text{ even}} I(\widetilde{X}\widetilde{Y} : C_t|XA'_t)_{\Psi^t}.$$

The Holevo quantum information complexity is defined as

$$\begin{aligned}\mathrm{HQIC}(\Pi,\mu) &= \sum_{t \text{ odd}} I(X : B'_tC_t|Y)_{\Psi^t} + \sum_{t \text{ even}} I(Y : A'_tC_t|X)_{\Psi^t} \\ &= \sum_{t \text{ odd}} I(X : B_tC_t|Y)_{\Psi^t} + \sum_{t \text{ even}} I(Y : A_tC_t|X)_{\Psi^t} \quad \text{(for safe protocols)}.\end{aligned}$$

For brevity, we shall often only use the classical input distribution μ as the subscript, or drop the subscript entirely, for these information quantities.

It was proved in [14], that for an r-round protocol Π^r, HQIC and QIC satisfy the following relation:

$$\mathrm{QIC}(\Pi^r,\mu) \geq \frac{1}{r}\,\mathrm{HQIC}(\Pi^r,\mu) \geq \frac{1}{2r}\,\mathrm{QIC}(\Pi^r,\mu).$$

Moreover, for any μ, $\mathrm{QIC}(\Pi,\mu) \leq \mathrm{QCC}(\Pi)$.

3 Lifting the classical adversary

3.1 The gadget and its properties

The gadget we use is the same one used in [10], called VER in that work. This is the function $\mathrm{VER}\colon \{0,1\}^2 \times \{0,1\}^2 \to \{0,1\}$ defined by $G(x,y) = 1$ if and only if $x+y$ is equivalent to 2 or 3 modulo 4, where $x, y \in \{0,1\}^2$ are interpreted as binary representations of integers in $\{0,1,2,3\}$. This gadget has the property of being *versatile*, which means that it satisfies the following three properties:

1. Flippability: given any input (x,y), Alice and Bob can perform a local operation on their respective inputs (without communicating) to get (x',y') such that $G(x',y') = 1 - G(x,y)$.

2. Random self-reducibility: given any input (x,y), Alice and Bob can use shared randomness (without communicating) to generate (x',y') which is uniformly distributed over $G^{-1}(G(x,y))$. That is, Alice and Bob can convert any 0-input into a random 0-input and any 1-input into a random 1-input, without any communication. More formally, if the domain of G is $\mathcal{X}\times\mathcal{Y}$, we require a probability distribution ν_G over pairs of permutations in $S_\mathcal{X} \times S_\mathcal{Y}$ such that for each $(x,y) \in \mathcal{X}\times\mathcal{Y}$, sampling (σ_A,σ_B) from ν_G and constructing the pair $(\sigma_A(x),\sigma_B(y))$ gives the uniform distribution over $G^{-1}(G(x,y))$.

3. Non-triviality: the function G contains AND_2 as a sub-function (and by flippability, it also contains OR_2 as a sub-function).

These three properties were established in [10] for the function VER; a gadget which satisfies them is called *versatile*. Our lifting proof will work for any versatile gadget G. We will need the following simple lemma, which allows us to generate n-tuples of inputs to G that evaluate to either a string $s \in \{0,1\}^n$ or its complement $\hat{s} \in \{0,1\}^n$. We use the notation $G^{-1}(s)$ to denote the set of all n-tuples of inputs to G that together evaluate to $s \in \{0,1\}^n$; this abuses notation slightly (we would technically need to write $(G^{\oplus n})^{-1}(s)$, where $G^{\oplus n}$ is the function we get by evaluating n independent inputs to G).

Lemma 9. *Let $s \in \{0,1\}^m$ be a given string and G be a versatile gadget. Then there is a protocol with no communication using shared randomness between Alice and Bob, who receive inputs (a,b) in the domain of G such that*

- *If $G(a,b) = 0$, Alice and Bob produce output strings (x,y) that are uniformly distributed in $G^{-1}(s)$*
- *If $G(a,b) = 1$, Alice and Bob produce output strings (x,y) that are uniformly distributed in $G^{-1}(\hat{s}) = G^{-1}(s \oplus 1^m)$.*

Proof. Alice and Bob share independent instances of the permutations ν_G, σ_A and σ_B as randomness. Applying independent instances of ν_G, Alice and Bob can produce (x', y') that are uniformly distributed in $G^{-1}((G(a,b))^m)$: this is done by applying m independent instances of σ_A and σ_B from ν_G to a and b respectively. Now Alice and Bob know where s differs from 0^m. By applying independent instances of the local flipping operation on x' and y' at these locations, they can negate the output of G. It is clear the resultant string (x,y) is uniformly distributed in $G^{-1}(s)$ if $G(a,b) = 0$ and in $G^{-1}(\hat{s})$ if $G(a,b) = 1$. □

We additionally have the following lemma, which uses the non-triviality property of a versatile gadget.

Lemma 10. *If G is a constant-sized non-trivial gadget (containing AND_2 and OR_2 as subfunctions), and μ_0 and μ_1 are uniform distributions over its 0- and 1-inputs, then any classical protocol Π for computing G with bounded error has $\mathrm{IC}(\Pi, \mu_0), \mathrm{IC}(\Pi, \mu_1) = \Omega(1)$.*

Proof. G contains the AND_2 function, and μ_0 puts uniform $\Omega(1)$ weight on the 0-inputs of the AND_2 subfunction. [8] showed that any protocol computing the AND_2 function must have $\Omega(1)$ information cost with respect to the distribution that puts $1/3$ weight on all 0-inputs of the AND_2 function. Hence any protocol for G must also have $\mathrm{IC}(\Pi, \mu_0) = \Omega(1)$. Similarly, by considering the fact that G contains the OR_2 function, we can show that $\mathrm{IC}(\Pi, \mu_1) = \Omega(1)$. □

Although we only need a single versatile gadget, such as VER, we will briefly remark that there is actually an infinite family of versatile gadgets, and that this family is universal (i.e. every communication function is a sub-function of some gadget in the family).

Lemma 11. *There is a universal family of versatile gadgets.*

Proof. For ease of notation, let G denote VER. For each $n \in \mathbb{N}^+$, we define G_n to be $\textsc{Parity}_n \circ G$. Note that G_n has the signature $\{0,1\}^{2n} \times \{0,1\}^{2n} \to \{0,1\}$. We observe that G_n is versatile for each $n \in \mathbb{N}^+$. This is because, given a single

input $((x_1, x_2, \ldots, x_n), (y_1, y_2, \ldots, y_n))$ to G_n with $x_i, y_i \in \{0, 1, 2, 3\}$ for each $i \in [n]$, Alice and Bob can locally generate the uniform distribution over all inputs with the same G_n-value. They can do this by first negating a random subset of the positions i of even size (using the flippability property of VER), and then converting each of the n resulting inputs to G into a random input to G with the same G-value.

Suppose z is the n-bit string with $z_i = G(x_i, y_i)$. Then flipping a random even subset of the bits of z is equivalent to generating a random string w that has the same parity as z. It follows that the above procedure generates a random input to G_n that has the same G_n value as the original input, meaning that G_n is random self-reducible. By flipping any single gadget G within G_n, we can negate G_n, so it is also flippable. Finally, since G_n contains G as a sub-function, it also contains AND as a sub-function, so G_n is versatile for each $n \in \mathbb{N}^+$.

It remains to show that $\{G_n\}_n$ is universal. We note that since G contains AND as a sub-function, and since $G_n = \textsc{Parity}_n \circ G$, the function G_n contains $\textsc{Parity}_n \circ \text{AND}$ as a sub-function. The latter is the inner product function IP_n on n bits, which is well-known to be universal. Hence G_n is also universal. □

3.2 The lifting theorem

Theorem 12. *Let G be a constant-sized versatile gadget such as VER, and let $f\colon \{0,1\}^n \to \Sigma$ be a relation. Then* $\text{RCC}(f \circ G) = \Omega(\text{CAdv}(f))$.

Proof. Let Π be a randomized protocol for $f \circ G$ which uses T rounds of communication (with one bit sent each round), and successfully computes $f \circ G$ with probability at least $1 - \epsilon/2$ for each input. Consider inputs XY distributed according to $\mu_z = \mu_{z_1} \otimes \ldots \otimes \mu_{z_n}$, where each μ_{z_i} is the uniform distribution over (x_i, y_i) in $G^{-1}(z_i)$. Suppose Π uses public randomness R which is independent of the inputs XY. We introduce the dependency-breaking random variables D and U [8] in the following way: D is independent of X, Y, R and is uniformly distributed on $\{0,1\}^n$. For each $i \in [n]$, if $D_i = 0$, then $U_i = X_i$, and if $D_i = 1$, then $U_i = Y_i$. Defined this way, given $D_i U_i$, X_i and Y_i are independent under μ_z. We shall use this algorithm to give a weight scheme $q'(z, i)$:

$$q'(z,i) = I(X_i : \Pi | X_{<i} Y D U R)_{\mu_z} + I(X_i : \Pi | Y_{<i} X D U R)_{\mu_z}$$

where $X_{<i}$ denotes $X_1 \ldots X_{i-1}$, and similarly for $Y_{<i}$. Clearly q is non-negative, and we shall show that

$$\sum_{i : z_i \neq w_i} \min\{q'(z,i), q'(w,i)\} = \Omega(1)$$

for all z, w such that $f(z) \cap f(w) = \varnothing$, where the constant in the $\Omega(1)$ is universal. Using this constant to normalize $q'(z, i)$, we get $q(z, i)$ which is a valid weight scheme. Since for any fixed value of $DU = du$, $I(X : \Pi | YR)_{\mu_{zdu}}$ is an information cost,

$$\sum_{i \in [n]} q'(z,i) = I(X : \Pi | Y D U R)_{\mu_z} + I(Y : \Pi | X D U R)_{\mu_z} \leq \text{CC}(\Pi),$$

we have for any protocol Π,

$$\text{CC}(\Pi) \geq \Omega\left(\sum_{i \in [n]} q(z,i)\right) \geq \Omega\left(\min_{\{q(z,i)\}} \sum_{i \in [n]} q(z,i)\right)$$

where the minimization is over all valid weight schemes. This proves the result.

Let z and w be two inputs to f such that $f(z) \cap f(w) = \varnothing$. Suppose z and w differ on indices in the block $\mathcal{B}$. Let $\mathcal{B}^1$ be the subset of indices in $\mathcal{B}$ where $\min\{q'(z,i), q'(w,i)\}$ is achieved by $q'(z,i)$, and $\mathcal{B}^2$ be the subset where the minimum is achieved by $q'(w,i)$. For an index $i \in \mathcal{B}^1$, let $\mathcal{B}^1_i$ denote $\mathcal{B}^1 \cap [i-1]$, and $\mathcal{B}^2_i$ denote $\mathcal{B}^2 \cap [i-1]$. We also use $\mathcal{B}^{1,c}$ to denote $[n] \setminus \mathcal{B}^1$, and $\mathcal{B}^{1,c}_i$ to denote $[i-1] \setminus \mathcal{B}^1_i$. Then,

$$
\begin{aligned}
&\sum_{i:z_i \neq w_i} \min\{q'(z,i), q'(w,i)\} \\
&= \sum_{i \in \mathcal{B}^1} (I(X_i : \Pi | X_{<i} Y D U R)_{\mu_z} + I(Y_i : \Pi | Y_{<i} X D U R)_{\mu_z}) \\
&\quad + \sum_{i \in \mathcal{B}^2} (I(X_i : \Pi | X_{<i} Y D U R)_{\mu_w} + I(Y_i : \Pi | Y_{<i} X D U R)_{\mu_w}) \\
&\overset{(1)}{=} \frac{1}{2} \sum_{i \in \mathcal{B}^1} (I(X_i : \Pi | X_{<i} Y D_{-i} U_{-i} R)_{\mu_z} + I(Y_i : \Pi | Y_{<i} X D_{-i} U_{-i} R)_{\mu_z}) \\
&\quad + \frac{1}{2} \sum_{i \in \mathcal{B}^2} (I(X_i : \Pi | X_{<i} Y D_{-i} U_{-i} R)_{\mu_w} + I(Y_i : \Pi | Y_{<i} X D_{-i} U_{-i} R)_{\mu_w}) \\
&\overset{(2)}{\geq} \frac{1}{2} \sum_{i \in \mathcal{B}^1} (I(X_i : \Pi | X_{\mathcal{B}^1_i} Y_{\mathcal{B}^1} D_{\mathcal{B}^{1,c}} U_{\mathcal{B}^{1,c}} R)_{\mu_z} + I(Y_i : \Pi | Y_{\mathcal{B}^1_i} X_{\mathcal{B}^1} D_{\mathcal{B}^{1,c}} U_{\mathcal{B}^{1,c}} R)_{\mu_z}) \\
&\quad + \frac{1}{2} \sum_{i \in \mathcal{B}^2} (I(X_i : \Pi | X_{\mathcal{B}^2_i} Y_{\mathcal{B}^2} D_{\mathcal{B}^{2,c}} U_{\mathcal{B}^{2,c}} R)_{\mu_w} + I(Y_i : \Pi | Y_{\mathcal{B}^2_i} X_{\mathcal{B}^2} D_{\mathcal{B}^{2,c}} U_{\mathcal{B}^{2,c}} R)_{\mu_w}) \\
&= \frac{1}{2}(I(X_{\mathcal{B}^1} : \Pi | Y_{\mathcal{B}^1} D_{\mathcal{B}^{1,c}} U_{\mathcal{B}^{1,c}} R)_{\mu_z} + I(Y_{\mathcal{B}^1} : \Pi | X_{\mathcal{B}^1} D_{\mathcal{B}^{1,c}} U_{\mathcal{B}^{2,c}} R)_{\mu_z}) \\
&\quad + \frac{1}{2}(I(X_{\mathcal{B}^2} : \Pi | Y_{\mathcal{B}^2} D_{\mathcal{B}^{2,c}} U_{\mathcal{B}^{2,c}} R)_{\mu_w} + I(Y_{\mathcal{B}^2} : \Pi | X_{\mathcal{B}^2} D_{\mathcal{B}^{2,c}} U_{\mathcal{B}^{2,c}} R)_{\mu_w}). \qquad (1)
\end{aligned}
$$

Above, equality (1) follows by using the definition of $D_i U_i$. The inequality (2) follows from the fact that given Y_i, X_i is independent of all other X_j-s, Y_j-s D_j-s and U_j-s under both the z and w distributions, hence $I(Y_i : \Pi | Y_{<i} X (DU)_{-i} R)_{\mu_z} \geq I(Y_i : \Pi | Y_{\mathcal{B}^1_i} X_{\mathcal{B}^1} (DU)_{\mathcal{B}^{1,c}} R)_{\mu_z}$, and equivalent inequalities hold for the other terms.

Consider $v \in \{0,1\}^n$ which agrees with w on the bits in $\mathcal{B}^1$, with z on the bits in $\mathcal{B}^2$, and with both of them outside $\mathcal{B}$. Since $f(z)$ and $f(w)$ are disjoint, at least one of the following must be true:

1. $\Pr_{(x,y)\sim\mu_v}[\Pi(x,y) \in f(z)] \leq \frac{1}{2}$
2. $\Pr_{(x,y)\sim\mu_v}[\Pi(x,y) \in f(w)] \leq \frac{1}{2}$.

In case 1, we shall give a protocol Π' that computes G correctly with probability at least $1-\epsilon$ in the worst case, such that

$$\mathrm{IC}(\Pi', \mu_0) = O(I(X_{\mathcal{B}^1} : \Pi | Y_{\mathcal{B}^1} D_{\mathcal{B}^{1,c}} U_{\mathcal{B}^{1,c}} R)_{\mu_z} + I(Y_{\mathcal{B}^1} : \Pi | X_{\mathcal{B}^1} D_{\mathcal{B}^{1,c}} U_{\mathcal{B}^{1,c}} R)_{\mu_z}).$$

Similarly, in case 2, we can use Π to give a protocol Π'' for G, such that

$$\mathrm{IC}(\Pi'', \mu_1) = O(I(X_{\mathcal{B}^2} : \Pi | Y_{\mathcal{B}^2} D_{\mathcal{B}^{2,c}} U_{\mathcal{B}^{2,c}} R)_{\mu_w} + I(Y_{\mathcal{B}^2} : \Pi | X_{\mathcal{B}^2} D_{\mathcal{B}^{2,c}} U_{\mathcal{B}^{2,c}} R)_{\mu_w}).$$

Due to equation (1) and Lemma 10, this proves the theorem.

In fact we only show how to construct the protocol Π' in case 1; the construction of Π'' is identical. Since z is in the domain of f and Π has worst case correctness for $f \circ G$, we must have $\Pr_{(x,y)\sim\mu_z}[\Pi(x,y) \in f(z)] \geq 1 - \epsilon/2$. Therefore, in case 1, Π can distinguish between samples from μ_z and μ_v on average: on getting a sample from μ_z or μ_v, we can run Π to see if it gives an output in $f(z)$ or not, and output z or v accordingly. This average distinguishing probability can be boosted by running Π multiple times.

In Π', Alice and Bob will share $RD_{\mathcal{B}^{1,c}}U_{\mathcal{B}^{1,c}}R_AR_B$ as randomness, where we use R_A and R_B to denote Alice and Bob's part of the shared randomness from Lemma 9, required to generate $z_{\mathcal{B}^1}$ if $G(a,b) = 0$ and $v_{\mathcal{B}^1}$ if $G(a,b) = 1$. On input (a,b) to G, Alice and Bob do the following k times for $k = \frac{2}{\epsilon^2}\ln(1/\epsilon)$:

- Alice sets $x_{\mathcal{B}^1} = R_A(a)$ and Bob sets $y_{\mathcal{B}^1} = R_B(b)$.
- Alice samples $x_{\mathcal{B}^{1,c}}$ and Bob samples $y_{\mathcal{B}^{1,c}}$ from private randomness, so that $G^{|\mathcal{B}^{1,c}|}(x_{\mathcal{B}^{1,c}}, y_{\mathcal{B}^{1,c}}) = z_{\mathcal{B}^{1,c}}$. They can do this since given $(DU)_{\mathcal{B}^{1,c}}$, $X_{\mathcal{B}^{1,c}}$ and $Y_{\mathcal{B}^{1,c}}$ are independent under μ_z and μ_v.
- They run Π on this (x,y) and generate the corresponding output.

(There are k independent instances of the D, U, R_A, R_B variables for each run above, but we denote all of them the same way for brevity.) The final output of Π' is 1 if the number of runs which have given an output in $f(z)$ is at least $(1-\epsilon)k$, and 0 otherwise.

Clearly if $G(a,b) = 0$, then (x,y) generated this way is uniformly distributed in the support of μ_z, and if $G(a,b) = 1$, then (x,y) is uniform in the support of μ_v. Calling the protocol in the i-th round of Π', Π_i, notice that the transcript of each Π_i is independent of AR_A, where A is the random variable for Alice's input, given the generated $X_{\mathcal{B}^1}$ (and this holds true even conditioned on $BR_BD_{\mathcal{B}^{1,c}}U_{\mathcal{B}^{1,c}}R$). Moreover, both $X_{\mathcal{B}^1}$ and Π_i are independent of BR_B given $Y_{\mathcal{B}^1}$ and of $Y_{\mathcal{B}^1}$ given BR_B (even conditioned on $D_{\mathcal{B}^1}U_{\mathcal{B}^1}R$). Let $\mu_{0,z}$ denote the distribution of $ABR_AR_B(DU)_{\mathcal{B}^{1,c}}$ when $G(a,b) = 0$, which induces the distribution $\mu_{z_{\mathcal{B}^1}}$ on $X_{\mathcal{B}^1}Y_{\mathcal{B}^1}$. Hence,

$$\begin{aligned}
I(A : \Pi_i | BR_AR_B(DU)_{\mathcal{B}^1}R)_{\mu_{0,z}} &\leq I(AR_A : \Pi_i | BR_B(DU)_{\mathcal{B}^{1,c}}R)_{\mu_{0,z}} \\
&\overset{(1)}{\leq} I(X_{\mathcal{B}^1} : \Pi_i | BR_B(DU)_{\mathcal{B}^{1,c}}R)_{\mu_{0,z}} \\
&\overset{(2)}{\leq} I(X_{\mathcal{B}^1} : \Pi_i | Y_{\mathcal{B}^1}BR_B(DU)_{\mathcal{B}^{1,c}}R)_{\mu_{0,z}} \\
&\overset{(3)}{=} I(X_{\mathcal{B}^1} : \Pi_i | Y_{\mathcal{B}^1}(DU)_{\mathcal{B}^{1,c}}R)_{\mu_{0,z}} \\
&\overset{(4)}{=} I(X_{\mathcal{B}^1} : \Pi | Y_{\mathcal{B}^1}(DU)_{\mathcal{B}^{1,c}}R)_{\mu_z}.
\end{aligned}$$

where inequality (1) follows from the fact that $I(U' : V|W) \leq I(U : V|W)$ if U' is independent of V given UW, (2) follows from the fact that $I(U : V|W) \leq I(U : V|WW')$ if V is independent of W' given W, and (3) follows from $I(U : V|WW') = I(U : V|W')$ if U and V are both independent of W given W'. Finally, (4) follows from the definition of Π_i and the the fact that the variables AB don't appear in the expression, so we can switch from $\mu_{0,z}$ to μ_z. Similarly,

$$I(B : \Pi_i | AR_AR_B(DU)_{\mathcal{B}^{1,c}}R)_{\mu_{0,z_{\mathcal{B}^1}}} \leq I(Y_{\mathcal{B}^1} : \Pi | X_{\mathcal{B}^1}(DU)_{\mathcal{B}^{1,c}}R)_{\mu_{z_{\mathcal{B}^1}}},$$

which lets us conclude that

$$\mathrm{IC}(\Pi_i, \mu_0) \leq I(X_{\mathcal{B}^1} : \Pi | Y_{\mathcal{B}^1}(DU)_{\mathcal{B}^{1,c}} R)_{\mu_z} + I(Y_{\mathcal{B}^1} : \Pi | X_{\mathcal{B}^1}(DU)_{\mathcal{B}^{1,c}} R)_{\mu_z}.$$

The final $\mathrm{IC}(\Pi', \mu_0)$ is then at most $k(I(X_{\mathcal{B}^1} : \Pi | Y_{\mathcal{B}^1}(DU)_{\mathcal{B}^{1,c}} R)_{\mu_z} + I(Y_{\mathcal{B}^1} : \Pi | X_{\mathcal{B}^1}(DU)_{\mathcal{B}^{1,c}} R)_{\mu_z})$.

Now let us analyze the worst case error made by Π'. Since the output of Π_i on (a, b) is expected output of Π on (x, y) uniformly sampled from either μ_z or μ_v, Π_i produces an output in $f(z)$ on (a, b) such that $G(a, b) = 0$ with probability at least $1 - \epsilon/2$, and on (a, b) such that $G(a, b) = 1$ with probability at most $\frac{1}{2}$. Hence by the Hoeffding bound, the probability of $(1 - \epsilon)k$ many 0 outputs in the first case is at least

$$1 - e^{-\epsilon^2 k/2} \geq 1 - \epsilon$$

and in the second case is at most

$$e^{-2(1/2-\epsilon)^2 k} \leq \epsilon.$$

Hence the probability of error on either input is at most ϵ. □

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